

**PUNE INSTITUTE OF COMPUTER TECHNOLOGY
DHANKAWADI, PUNE – 43.**

LIST OF LAB EXPERIMENTS

ACADEMIC YEAR: 2026-27

Date: 6/07/2026

DEPARTMENT: INFORMATION TECHNOLOGY

CLASS: S.Y. B. TECH.

SEMESTER: III

SUBJECT: DATA STRUCTURES AND APPLICATIONS LABORATORY (DSAL)

LAB EXPT. NO	PROBLEM STATEMENT
Practice Assignments	
1.	A. Implement Singly Linked List (SLL) to perform basic operations such as Insertion, Deletion, Traversal, Searching, updating node value, and reversing the list. B. Develop a Contact Management System using SLL that allows users to add, display, search, update, and delete contact records. The system should store details such as name, phone number, email address, and other relevant information, and provide an easy-to-use menu-driven interface for managing contacts efficiently.
2.	A. Implement Doubly Linked List (DLL) to perform basic operations such as Insertion, Deletion, Traversal, Searching, updating node value, and reversing the list. B. Develop a Music Playlist Manager that allows user to add songs, delete songs, search for a song, display the playlist and navigate through songs in both forward (Next) and backward (Previous) direction using menu driven interface.
3.	A. Implement the Stack using an Array and Linked List to perform basic operations such as Push, Pop, Peek (Top), Display and Check whether Stack is Empty or Full. B. Implement a program to convert an infix expression to prefix and postfix notation. Evaluate both prefix and postfix expressions.
4.	A. Implement Queue using an Array and Linked List to perform basic operations such as Enqueue, Dequeue, Front, Display, and Check whether Queue is Empty or Full. B. Develop a Hospital Emergency Room Management System using a Priority Queue. The system should allow users to register patients along with their priority level (e.g., Critical, Serious, Normal), display the waiting queue, and serve patients based on priority basis.
5.	A. Implement Linear Search and Binary Search Algorithm to search an element in the list B. Develop a Student Record Management System that stores student details and searches for a student by roll number or name using the Linear Search algorithm. The program should display the student's details if found; otherwise, it should display an appropriate message. C. Develop a Library Book Search System that stores book records sorted by Book ID and searches for a book using the Binary Search algorithm. The program should display the book details if found; otherwise, it should display an appropriate message.

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6.	A. Implement Bubble Sort and Insertion Sort Algorithm to sort a list of elements in ascending order. B. Develop a Student Marks Management System that stores the marks of students (along with other details) and sorts them in ascending order using Insertion Sort and in descending order using Bubble Sort.
7.	A. Implement Quick Sort Algorithm to sort a list of elements in ascending order using divide and conquer approach. B. Develop a Product Management System that stores product details such as Product ID, Product Name, and Price, and sorts the products based on their price using the Quick Sort algorithm. The program should display the sorted list in ascending or descending order through a menu-driven interface.
8.	A. Implement Hash Table using open addressing to perform insertion and search of elements using an appropriate probing technique (Linear and Quadratic). B. Implement Hash Table using separate chaining to perform insertion and search of elements.
9.	Implement sequential file to store records, display all records, search for a specific record, delete a specific record, and append new records while maintaining sequential access.
10.	"Real-World Solution Development Using Fundamental Data Structures and Algorithms" Problem Statement: Design and develop a real-world application that leverages fundamental concepts of Data structures and Algorithms (DSA), including linear data structures, searching, and sorting techniques (with a preference for hashing). The solution should efficiently handle data storage, retrieval, and processing while optimizing performance.

**Head of Department
(Information Technology)**

**Subject Coordinator
(Sachin D. Shelke)**